Causal Inference with Ordinal Outcomes: Flexible Estimation Framework with Normalizing Flows

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Abstract

Ordinal outcomes, such as Likert-type survey responses, are ubiquitous in political science, yet they pose distinctive challenges for causal inference. These outcomes are routinely analyzed by assigning numerical scores and applying linear models, despite longstanding concerns that ordinal responses do not identify treatment effects on an underlying cardinal scale without additional assumptions. Building on the classical identification result for ordinal outcomes and existing recommendations to report probability-based quantities, I develop a flexible likelihood-based framework for estimating distributional causal effects, including category-specific and cumulative treatment effects, which are invariant to admissible transformations of the latent scale. Existing approaches typically estimate these quantities using parametric ordered logit or ordered probit models, which rely on restrictive assumptions about the latent error distribution and index specification. This paper proposes a flexible likelihood-based estimation framework for ordinal causal inference using conditional normalizing flows. The proposed estimator flexibly learns the conditional distribution underlying observed ordinal responses without imposing a parametric latent error distribution, while preserving the likelihood-based structure familiar from classical ordered response models. The same framework accommodates both structured latent-index models and fully nonparametric conditional distribution estimators, allowing researchers to balance interpretability and modeling flexibility within a unified estimation procedure. Monte Carlo simulations show that the proposed estimator maintains low bias across a broad range of latent error distributions and nonlinear response mechanisms under which conventional ordered response models perform poorly. Replications of two published survey experiments demonstrate that flexible distributional estimation can materially alter substantive conclusions about experimental treatment effects.

1. Introduction

Causal inference has become a central organizing principle in political science. Experimental and quasi-experimental designs, including survey experiments, difference-in-differences, and regression discontinuity designs, are now standard empirical strategies for studying the effects of policies, institutions, and information. Survey experiments are particularly prominent because random assignment provides a transparent basis for causal identification. Between 2021 and 2024, roughly 20% of articles published in the *American Political Science Review*, *American Journal of Political Science*, and *Journal of Politics* employed a survey experiment as part of their empirical strategy.

Many of these studies investigate outcomes that are inherently latent, including support for redistribution (Alt and Iversen, 2017; Magni, 2021), evaluations of political leaders (Canes-Wrone and De Marchi, 2002; Kriner and Schwartz, 2009), and foreign policy preferences (Tomz and Weeks, 2020). Although these concepts are commonly viewed as lying on an underlying continuous attitude scale, they are almost always observed through ordinal response categories such as Likert-type scales ranging from *strongly disagree* to *strongly agree*. Consequently, causal inference in political science routinely relies on outcomes that preserve the ordering of responses but not the distances between them.

Applied researchers typically analyze ordinal outcomes in one of three ways. The most common approach assigns numerical scores to response categories and estimates treatment effects using difference-in-means or ordinary least squares, implicitly treating ordinal responses as cardinal. Another common strategy collapses multiple categories into a binary outcome before estimation, sacrificing information about intermediate responses. A third approach fits ordered response models, such as ordered logit or ordered probit, which explicitly account for the ordinal nature of the data through a latent-variable representation. Each approach reflects a different set of assumptions about the relationship between the observed ordinal responses and the underlying latent attitude.

This paper begins from a classical observation in the ordinal response literature: latent treatment effects are identified only up to location and scale transformations of the underlying latent variable. Consequently, treatment effects defined through arbitrary numerical codings of ordinal responses generally require additional measurement assumptions linking the observed categories to a cardinal latent scale (Schröder and Yitzhaki, 2017; Bond and Lang, 2018; Bloem, 2022). This observation motivates a focus on distributional causal quantities, such as category-specific and cumulative treatment effects, which are defined directly on the observed ordinal responses and do not rely on assumptions about the cardinal spacing between adjacent categories.

This perspective complements recent recommendations in political methodology to summarize ordinal treatment effects using predicted probabilities rather than latent coefficients (Hanmer and Ozan Kalkan, 2013). Rather than revisiting how ordered response models should be interpreted, this paper asks how these probability-based causal quantities should be estimated when the assumptions underlying conventional ordered response models are difficult to justify. The methodological focus therefore shifts from interpreting latent coefficients to flexibly estimating the conditional ordinal response distribution. Existing semiparametric estimators relax some of these assumptions but remain difficult to apply with rich covariate structures or nonlinear response mechanisms that are common in modern survey research.

Estimating these distributional causal effects requires modeling the conditional distribution of ordinal responses. Ordered logit and ordered probit remain the workhorses of applied political science because they efficiently exploit the ordering of response categories, admit interpretable latent-variable representations, and often perform well when their assumptions are approximately satisfied. Their validity, however, depends on assumptions about the latent error distribution and index specification that may be difficult to justify in many survey experiments. Political attitudes are frequently polarized or clustered, producing multimodal latent distributions, while persuasive treatments often affect moderate respondents differently from strong supporters, generating nonlinear treatment-response relationships. In such settings, misspecification can distort estimated category probabilities and cumulative treatment effects, even when the experimental design itself identifies the causal effect. The practical challenge is therefore not causal identification, but flexible estimation of the conditional ordinal response distribution.

This paper develops a flexible likelihood-based framework for estimating causal effects with ordinal outcomes. The proposed estimator employs conditional normalizing flows to model the conditional response distribution without imposing a fixed parametric latent error distribution. Unlike existing semiparametric estimators, the framework remains computationally feasible with high-dimensional covariates while preserving exact likelihood-based inference. A single estimation procedure spans classical latent-index models, generalized latent-index specifications, and fully nonparametric conditional response distributions, allowing researchers to balance structural assumptions and modeling flexibility while targeting the same probability-based causal quantities.

I evaluate the proposed framework through Monte Carlo simulations designed to reflect empirically realistic survey settings, including polarized latent attitudes, nonlinear treatment-response relationships, and non-Gaussian latent error distributions. Across these settings, the proposed estimator maintains low bias while conventional ordered response models exhibit substantial distortions in estimated probability effects. I then revisit the survey experiments of Tomz and Weeks (2020) and Mattingly et al. (2025), showing that flexible estimation of the ordinal response distribution can materially alter substantive conclusions regarding the magnitude and distribution of treatment effects.

The remainder of the paper proceeds as follows. Section 2 formalizes causal inference with ordinal outcomes and motivates distributional causal estimands. Section 3 reviews existing estimation approaches and introduces the proposed likelihood-based framework. Section 4 evaluates finite-sample performance through Monte Carlo simulations. Section 5 presents the empirical applications. Section 7 concludes.

2. Causal Inference with Ordinal Outcomes

This section develops the conceptual framework underlying the remainder of the paper. I begin by describing ordinal survey responses as measurements of an underlying latent attitude using the classical threshold-crossing model. I then revisit a well-known identification result showing that latent treatment effects are identified only up to location and scale, and discuss its implications for causal inference with ordinal outcomes. Finally, I introduce distributional causal estimands that are directly identified from observed ordinal responses and motivate the estimation framework developed in the next section.

2.1. Latent Outcomes, Ordinal Responses, and Causal Effects

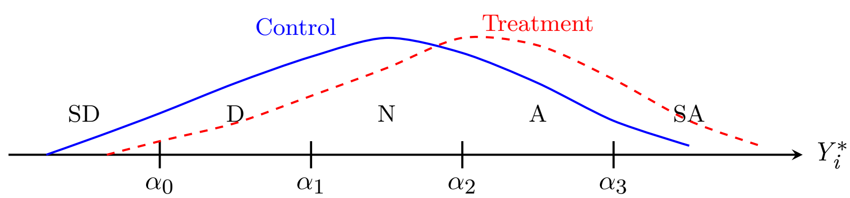

FIGURE 1. Ordinal responses arise by partitioning an underlying latent attitude into ordered response categories. The observed data identify how treatment changes probabilities of falling into response categories, but not the absolute scale of the latent attitude.

Many quantities of interest in political science, such as support for redistribution (Alt and Iversen, 2017; Magni, 2021), immigration preferences (Mayda and Rodrik, 2005), trust in institutions, and foreign policy attitudes (Scheve and Slaughter, 2001; Tomz and Weeks, 2020), are naturally viewed as continuous latent constructs. Survey respondents, however, rarely report these attitudes on a continuous scale. Instead, researchers observe ordinal responses, such as Likert items ranging from *strongly disagree* to *strongly agree*, ordered approval categories, or frequency scales. Consequently, empirical analyses are conducted using outcomes that preserve the ordering of respondents' attitudes but not the distances between adjacent response categories.

Following the classical ordered response framework, let Y_i^* denote respondent i's latent attitude and let $Y_i \in 0, 1, \ldots, J$ denote the observed ordinal response. The observed response is generated through a threshold-crossing mechanism,

$$Y_i = j \iff \alpha_{j-1} < Y_i^* \leq \alpha_j, \qquad j = 0, \ldots, J,$$

where

$$-\infty = \alpha_{-1} < \alpha_0 < \cdots < \alpha_J = \infty$$

are unknown thresholds partitioning the latent scale. Figure 1 illustrates this measurement process. Throughout the paper I assume respondents share a common reporting function. Section XX discusses extensions allowing reporting thresholds to vary across respondents following generalized ordered response models (Williams, 2006). Alternative survey designs, such as anchored vignettes (King et al., 2004; King and Wand, 2007), provide complementary approaches for addressing heterogeneous reporting behavior.

The threshold-crossing representation should be interpreted as a measurement model rather than a structural model. It formalizes the idea that ordinal responses preserve the ordering of the underlying attitudes while discarding information about the distances between adjacent categories. Consequently, the observed responses provide only partial information about the latent construct: they reveal which interval of the latent scale a respondent occupies, but not the respondent's exact location within that interval.

To describe causal effects, let $Y_i^*(d)$ denote the latent potential outcome under treatment level d, and let

$$Y_i(d) = g\left(Y_i^*(d)\right)$$

denote the corresponding observed ordinal potential outcome. Throughout the paper I adopt the standard assumptions of SUTVA, conditional ignorability, and positivity (Rubin, 1974). If the latent outcome Y_i^* were directly observed, these assumptions would suffice to identify conventional causal estimands such as

$$ATE^* = \mathbb{E} = \left[Y_i^*(1) - Y_i^*(0)\right].$$

In practice, however, researchers observe only the ordinal response Y_i. The central question is therefore not whether causal effects are identified, but which causal quantities remain identified once the underlying latent attitude is observed only through ordered response categories. The next subsection revisits the classical identification result for ordinal response models and uses it to distinguish between causal quantities that require additional measurement assumptions and those that do not.

2.2. Identification of Causal Effects from Ordinal Outcomes

The threshold-crossing framework provides a useful representation of ordinal outcomes, but it also reveals a fundamental limitation of what can be learned from ordinal data. A classical result in the ordered response literature is that latent-variable models are identified only up to location and scale. Although this normalization is standard in econometrics and psychometrics, its implications for causal inference have received comparatively little attention in applied political science. In particular, survey experiments frequently interpret treatment effects estimated from numerically coded ordinal responses as effects on an underlying latent attitude without making explicit the measurement assumptions required for such an interpretation.

To formalize this point, suppose the latent outcome satisfies

$$Y_i^* = h_\theta(X_i) + D_i^\top \tau + \varepsilon_i, \tag{1}$$

where $h(X_i, \beta)$ denotes the systematic component of the latent outcome, τ is the treatment effect on the latent scale, and ε_i has cumulative distribution function F. Combining this latent-index specification with the threshold-crossing measurement model implies

$$\begin{aligned} P(Y_i = j \mid X_i, D_i) &= F\left(\alpha_j - h(X_i, \beta) - D_i^\top \tau\right) \\ &= F\left(\alpha_{j-1} - h(X_i, \beta) - D_i^\top \tau\right). \end{aligned}$$

Now consider any constants $a \in \mathbb{R}$ and $c > 0$, and define a transformed latent outcome

$$\tilde{Y}_i^* = a + cY_i^*,$$

together with transformed thresholds

$$\tilde{\alpha}_j = a + c\alpha_j.$$

Because both the latent outcome and the thresholds are transformed simultaneously, the probabilities of the observed ordinal responses remain unchanged.

Theorem 1 (Location and Scale Nonidentification). *Under the threshold-crossing latent-variable model, any positive affine transformation of the latent outcome and thresholds generates the same distribution of the observed ordinal responses. Consequently, treatment effects on the latent outcome are identified only up to location and scale.*

Theorem 1 should not be interpreted as a failure of causal identification. Under the standard assumptions of SUTVA, conditional ignorability, and positivity, causal effects remain identified for any well-defined outcome. Rather, the theorem highlights a measurement problem. Ordinal responses preserve the ordering of latent attitudes but do not reveal the cardinal scale on which those attitudes are measured. Consequently, causal quantities defined on the latent scale require additional measurement assumptions beyond those needed for causal identification itself.

This distinction has immediate implications for empirical practice. Different analyses of ordinal outcomes target different causal quantities and therefore rely on different measurement assumptions. For example, assigning numerical scores to ordinal categories and estimating mean differences implicitly assumes that those scores provide a meaningful cardinal representation of the latent attitude. By contrast, causal quantities defined directly on the observed ordinal responses do not require assumptions about the cardinal spacing between adjacent categories. The next subsection formalizes these probability-based causal estimands and shows how they arise naturally from the observed ordinal response distribution.

2.3. Distributional Causal Estimands

The identification result in Theorem 1 does not imply that causal inference with ordinal outcomes is fundamentally limited. Rather, it suggests that inference should focus on causal quantities defined directly on the observed ordinal responses. Such quantities require no assumptions about the cardinal spacing of the latent attitude and therefore remain invariant to admissible transformations of the latent scale. This perspective is closely aligned with recommendations to report predicted probabilities rather than latent coefficients when interpreting ordinal response models (Hanmer and Ozan Kalkan, 2013). Here I formalize these quantities as causal estimands and show that they are identified under the standard assumptions of causal inference.

The fundamental object is the distribution of potential ordinal outcomes. Let

$$\pi_j(d) = P\left(Y_i(d) = j\right), \qquad j = 0, \ldots, J,$$

denote the probability that a randomly selected individual reports category j under treatment level d.

Under the standard assumptions of SUTVA, conditional ignorability, and positivity, these probabilities are identified by standardization,

$$\pi_j(d) = \mathbb{E}_X\left[P\left(Y_i = j \mid D_i = d, X_i\right)\right],$$

where the expectation is taken over the covariate distribution of the target population. In randomized experiments, this expression simplifies to the observed proportion of respondents falling into category j within each treatment arm.

Proposition 1. *Under SUTVA, conditional ignorability, and positivity, the distribution of potential ordinal outcomes is identified by standardization. Consequently, any causal quantity that is a functional of this distribution is also identified.*

This proposition has an important practical implication. Once the conditional ordinal response distribution has been estimated, a wide range of causal quantities become available without fitting additional models. The primary statistical task is therefore estimation of the conditional response probabilities,

$$P(Y_i = j \mid D_i, X_i),$$

which serve as the building blocks for all distributional causal estimands considered in this paper.

The simplest causal quantity is the category-specific treatment effect,

$$\Delta_j = \pi_j(1) - \pi_j(0),$$

which measures how treatment changes the probability that a respondent selects category j. For example, in a survey experiment on support for military intervention, $\Delta_{\text{Strongly Agree}}$ measures the change in the proportion of respondents expressing the strongest level of support.

In many political science applications, however, interest centers on whether respondents exceed a substantively meaningful threshold rather than on a single response category. This motivates cumulative treatment effects,

$$\Delta_{\geq j} = P\left(Y_i(1) \geq j\right) = P\left(Y_i(0) \geq j\right),$$

which quantify changes in the probability of responding at or above category j. For example, a researcher may wish to estimate whether an informational treatment increases the probability that respondents express at least moderate support for a policy. Unlike dichotomizing the outcome before estimation, cumulative treatment effects can be calculated for every threshold after estimating the ordinal response distribution, allowing the full information contained in the ordinal responses to be retained.

These estimands also clarify the relationship between the proposed framework and existing empirical practice. Assigning numerical scores to ordinal categories estimates

$$\mathbb{E} = \left[s(Y_i(1)) - s(Y_i(0))\right],$$

where $s(\cdot)$ denotes the chosen scoring rule. This quantity is well defined for a fixed coding rule but generally does not identify a treatment effect on the underlying latent attitude. Similarly, dichotomizing an ordinal outcome estimates one cumulative treatment effect corresponding to a single threshold while discarding information about the remaining response categories. Estimating the full ordinal response distribution instead makes both category-specific and cumulative treatment effects available simultaneously, allowing researchers to select the quantity most appropriate for their substantive question after estimation rather than before it.

The discussion above highlights an important feature of distributional causal inference. Rather than targeting a single summary measure, the primary object of interest is the conditional distribution of ordinal responses,

$$P(Y_i = j \mid D_i, X_i),$$

from which a wide range of causal quantities can be obtained. Category-specific treatment effects and cumulative treatment effects are the primary estimands considered in this paper because they are directly interpretable and commonly reported in applied political science. More generally, however, any functional of the estimated response distributions may be calculated after estimation, including global measures of distributional change such as the one-dimensional Wasserstein (earth mover's) distance. Consequently, the statistical problem addressed in the remainder of this paper is estimation of the conditional ordinal response distribution rather than any single causal summary.

3. Flexible Estimation of Ordinal Response Distributions

Section 2 established that the causal quantities of interest are functionals of the conditional ordinal response distribution,

$$P(Y_i = j \mid D_i, X_i).$$

The remaining statistical problem is therefore estimation of this distribution. This section first reviews existing estimation approaches and their underlying assumptions before introducing a flexible likelihood-based framework that generalizes classical ordered response models while preserving their probabilistic interpretation.

3.1. Existing Estimation Approaches

The dominant approach to modeling ordinal outcomes in political science is the ordered response model, most commonly ordered logit or ordered probit. These models combine the latent-index representation in Equation (1) with the threshold-crossing measurement model introduced in Section 2.1. The distinction between ordered probit and ordered logit lies only in the assumed distribution of the latent error term, with the former assuming a standard normal distribution and the latter assuming a standard logistic distribution.

Ordered response models remain the workhorses of applied political science because they naturally respect the ordinal structure of survey responses, provide interpretable latent-variable representations, and retain the familiar likelihood-based inferential framework. Moreover, political methodologists have long advocated interpreting these models through predicted probabilities and first differences rather than latent coefficients (Hanmer and Ozan Kalkan, 2013), making them well suited for estimating the distributional causal quantities introduced in Section 2.3.

The validity of these probability estimates, however, depends on the assumptions used to construct the conditional ordinal response distribution. Classical ordered response models impose two principal restrictions. First, the latent error distribution is assumed to belong to a fixed parametric family, typically normal or logistic. Second, the systematic component of the latent outcome is specified through a parametric single-index model. Together, these assumptions determine the shape of the conditional response distribution and therefore the estimated category probabilities.

These assumptions are often reasonable, and when approximately correct ordered response models are statistically efficient. In many survey experiments, however, they may be difficult to justify. Political attitudes are frequently polarized or clustered, producing multimodal latent distributions. Ceiling and floor effects compress responses near the boundaries of Likert scales. Persuasive treatments may affect moderate respondents differently from strong supporters, generating nonlinear treatment-response relationships. Under such conditions, misspecification of either the latent error distribution or the systematic component may distort the estimated category probabilities that form the basis of substantive interpretation.

A substantial literature has relaxed these assumptions. Generalized ordered response models allow threshold-specific effects, while semiparametric estimators relax the distributional assumption on the latent error distribution. Bayesian nonparametric approaches provide additional flexibility through hierarchical modeling. These methods represent important advances, but they typically relax only one component of the model or become computationally demanding in the presence of high-dimensional conditioning variables. The next subsection develops a unified framework that relaxes both sources of misspecification while preserving the likelihood-based structure of conventional ordered response models.

3.2. A Flexible Likelihood-Based Framework

The preceding discussion suggests that the objective is not simply to replace the normal or logistic error distribution with a more flexible alternative. Rather, the goal is to estimate the conditional ordinal response distribution while preserving the probabilistic structure that makes ordered response models attractive. An ideal estimator should satisfy four properties. It should remain likelihood-based, accommodate high-dimensional conditioning variables, recover the full conditional response distribution, and permit varying degrees of structural modeling according to substantive knowledge.

The proposed framework retains the latent outcome representation in Equation (1) but generalizes how its components are modeled. Recall that the latent outcome consists of two parts: the systematic component

$h(\cdot)$ and the latent error distribution F. Existing estimators correspond to particular choices of these two components. Ordered logit and ordered probit specify both parametrically, whereas semiparametric estimators typically relax only the latent error distribution.

The proposed framework allows flexibility in either or both components while retaining a common likelihood-based estimation procedure. When substantive theory suggests that treatment and covariates operate through a common latent index, the systematic component may be specified parametrically while the latent error distribution is estimated flexibly. This specification may be viewed as a semiparametric generalization of ordered response models. Alternatively, when the functional relationship between covariates and the latent outcome is itself uncertain, the systematic component may also be estimated nonparametrically. In this way, the same estimation framework spans classical latent-index models, semiparametric ordered response models, and fully flexible conditional distribution estimators without requiring different inferential procedures.

The framework is implemented using conditional normalizing flows to model unknown probability distributions. Unlike kernel-based semiparametric estimators, conditional normalizing flows scale naturally to high-dimensional conditioning variables while retaining exact likelihood evaluation. Consequently, they preserve the familiar inferential machinery of ordered response models, including maximum-likelihood estimation, likelihood-based model comparison, and probabilistic calibration, while substantially relaxing distributional assumptions.

A particularly attractive feature of ordinal response models is that the latent error is scalar. Combined with monotone normalizing flows, this permits exact evaluation of both the latent density and its cumulative distribution function. Consequently, the threshold-crossing likelihood can be evaluated exactly without numerical integration, allowing the proposed estimator to remain computationally tractable while flexibly modeling the latent error distribution.

3.3. Modeling the Latent Error Distribution with Normalizing Flows

The proposed framework replaces the fixed parametric latent error distribution with a flexible conditional density estimator. Rather than assuming that the latent error follows a standard normal or logistic distribution, I estimate its conditional distribution directly from the data while retaining exact likelihood-based inference.

Normalizing flows provide a natural framework for this task. Originally introduced by Rezende and Mohamed (2016), normalizing flows represent an unknown probability distribution through a sequence of invertible transformations of a simple base distribution. Because each transformation is invertible, probability densities remain analytically tractable through the change-of-variables formula, allowing estimation to pro-

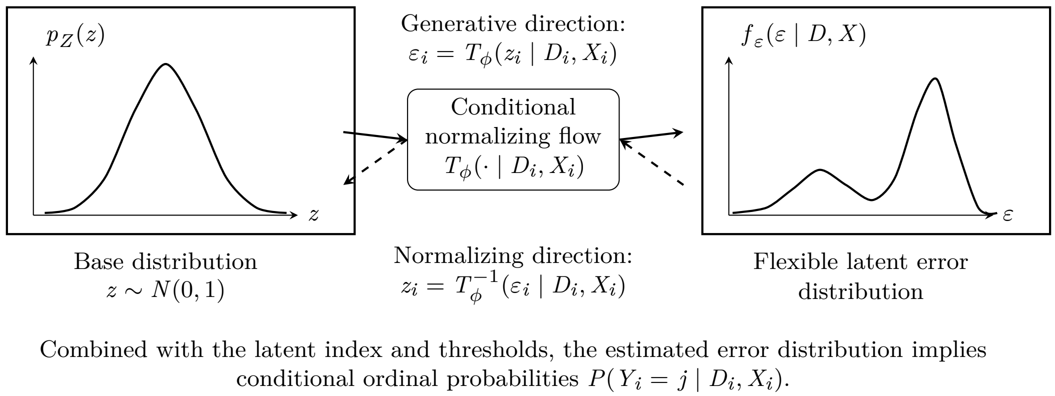

FIGURE 2. A conditional normalizing flow transforms a simple base distribution into a flexible latent error distribution. In the proposed ordered-response framework, this replaces the fixed normal or logistic error assumption while retaining likelihood-based estimation.

ceed by maximum likelihood. Since their introduction, normalizing flows have become a standard approach for flexible likelihood-based density estimation (Kobyzev, Prince and Brubaker, 2021; Papamakarios et al., 2021).

Figure 2 illustrates the basic idea. Let

$$z_i \sim p_Z(z)$$

denote a simple base distribution, taken throughout this paper to be the standard normal distribution. A conditional normalizing flow transforms this base distribution into the latent error distribution through an invertible mapping,

$$\varepsilon_i = T_\phi(z_i \mid D_i, X_i),$$

where the transformation is parameterized by ϕ and conditioned on the treatment and covariates. The same transformation may be inverted,

$$z_i = T_\phi^{-1}(\varepsilon_i \mid D_i, X_i),$$

allowing observations to be mapped back to the base distribution. This invertibility is the defining feature of normalizing flows and enables exact likelihood evaluation through the standard change-of-variables formula,

$$f_\varepsilon(\varepsilon_i \mid D_i, X_i) = p_Z(z_i) \left| \det \frac{\partial T_\phi^{-1}(\varepsilon_i \mid D_i, X_i)}{\partial \varepsilon_i} \right|.$$

Unlike kernel density estimators, the transformation is represented by a parametric function that scales naturally to high-dimensional conditioning variables. Throughout this paper I construct $T_\phi(\cdot)$ using monotonic rational-quadratic spline (RQS) transformations (Durkan et al., 2019). RQS flows provide smooth monotone mappings while retaining closed-form evaluation of both the inverse transformation and the Jacobian (Durkan et al., 2019). Because the latent error is one-dimensional, this construction also permits exact evaluation of the implied cumulative distribution function, which is required to compute the threshold probabilities in ordered response models.

The conditional flow is used only to model the latent error distribution. The systematic component of the latent outcome in Equation (1) remains independent of this choice and may therefore be specified at different levels of structural complexity. In the structured specification, the function $h(\cdot)$ is parameterized using a conventional latent-index model, so that flexibility is introduced only through the latent error distribution. This specification may be viewed as a semiparametric generalization of ordered logit and ordered probit. In the fully flexible specification, both the systematic component and the latent error distribution are modeled nonparametrically, allowing the conditional ordinal response distribution to be learned with minimal functional-form assumptions. Both specifications share the same likelihood and estimation procedure, differing only in the amount of structure imposed on the latent outcome model.

3.4. Likelihood-Based Estimation

The latent-index representation in Equation (1), together with the threshold-crossing measurement model introduced in Section 2.1, implies that the probability of observing response category j is

$$\begin{aligned} P(Y_i = j \mid D_i, X_i) &= P\left(\alpha_{j-1} < Y_i^* \leq \alpha_j \mid D_i, X_i\right) \\ &= F_\phi\left(\alpha_j - h(X_i) - D_i^\top \tau \mid D_i, X_i\right) - F_\phi\left(\alpha_{j-1} - h(X_i) - D_i^\top \tau \mid D_i, X_i\right), \end{aligned}$$

where $F_\phi(\cdot)$ denotes the conditional cumulative distribution function implied by the normalizing flow introduced in the previous subsection. This expression differs from ordered logit and ordered probit only through the choice of F_ϕ. Rather than assuming a logistic or normal distribution, the latent error distribution is estimated directly from the data.

Also, Unlike general multivariate flow models, the one-dimensional monotone spline construction used here admits exact evaluation of the conditional cumulative distribution function F_ϕ. Consequently, the category probabilities required by the ordered-response likelihood can be computed analytically without numerical integration.

Assuming independent observations, the log-likelihood for the observed ordinal responses is

$$\ell(\Theta) = \sum_{i=1}^{n} \sum_{j=0}^{J} 1(Y_i = j) \log P(Y_i = j \mid D_i, X_i),$$

where

$$\Theta = \tau, h, \phi, \alpha$$

collects all unknown parameters of the latent outcome model, the conditional normalizing flow, and the reporting thresholds.

The parameters are estimated by maximizing this likelihood using stochastic gradient optimization. Because the conditional normalizing flow admits exact evaluation of both the latent density and its cumulative distribution through the invertible transformation, the resulting objective remains a genuine likelihood rather than a surrogate prediction loss. Consequently, estimation retains many of the desirable features of conventional ordered response models, including likelihood-based model comparison and probabilistic calibration, while allowing substantially greater flexibility in the latent error distribution.

The likelihood in Equation (1) encompasses both structured and flexible specifications discussed in the previous subsection. When the systematic component $h(\cdot)$ is parameterized linearly, the estimator may be viewed as a semiparametric generalization of ordered logit and ordered probit. When $h(\cdot)$ is itself modeled nonparametrically, the same likelihood estimates the conditional ordinal response distribution under minimal functional-form assumptions. In either case, the fitted model directly yields the conditional probabilities

$$P(Y_i = j \mid D_i, X_i),$$

which serve as the basis for estimating all causal quantities introduced in Section 2.3. For example, the estimated category-specific treatment effect is obtained by standardization,

$$\widehat{\Delta}_j = \frac{1}{n} \sum_{i=1}^{n} \left[\widehat{P}(Y_i = j \mid D_i = 1, X_i) - \widehat{P}(Y_i = j \mid D_i = 0, X_i) \right],$$

which corresponds to the plug-in estimator of the g-formula (Robins, 1986). Cumulative treatment effects are computed analogously,

$$\widehat{\Delta}_{\geq j} = \frac{1}{n} \sum_{i=1}^{n} \left[\widehat{P}(Y_i \geq j \mid D_i = 1, X_i) - \widehat{P}(Y_i \geq j \mid D_i = 0, X_i) \right].$$

More generally, because the entire conditional response distribution is available after estimation, any functional of that distribution may be computed without fitting additional models. Examples include average predicted probabilities, first differences, stochastic dominance measures, or global summaries of distributional change such as the one-dimensional Wasserstein (earth mover's) distance. The proposed estimator therefore separates estimation from interpretation: the model is fitted once, while substantive quantities of interest are obtained as post-estimation summaries of the estimated response distribution.

The simulation study and empirical applications that follow focus primarily on category-specific and cumulative treatment effects because these quantities are directly interpretable and commonly reported in applied political science. The same fitted model, however, supports a substantially broader class of distributional summaries whenever alternative scientific questions arise. Throughout the simulations and empirical applications, uncertainty is quantified using the nonparametric bootstrap, which naturally propagates estimation uncertainty from the fitted conditional response distribution to all post-estimation functionals.

4. Monte Carlo Simulation

This section evaluates the finite-sample performance of the proposed estimation framework. The objective is not to demonstrate that flexible estimation universally outperforms classical ordered response models. Ordered logit and ordered probit remain statistically efficient when their assumptions are approximately satisfied. Instead, the simulations investigate the conditions under which flexible estimation of the conditional ordinal response distribution improves recovery of the distributional causal effects introduced in Section 2.3. Throughout, causal identification is held fixed so that differences in performance arise solely from the estimation procedure.

4.1. Simulation Design

Each simulation begins by generating a synthetic population of one million observations from the latent-index model in Equation (1). The latent outcome is converted into an observed ordinal response through the threshold-crossing measurement model introduced in Section 2.1, with common reporting thresholds fixed across treatment conditions. The resulting population serves as the ground truth throughout the Monte Carlo experiment. Because the population is sufficiently large, the true category-specific and cumulative treatment effects can be computed with negligible Monte Carlo error before sampling begins. Consequently, all reported estimation error reflects finite-sample performance rather than uncertainty in the reference values.

Treatment assignment is completely randomized, ensuring that the identification assumptions developed in Section 2 hold by construction. For each data-generating process, I draw 100 independent random samples of sizes $n \in 500, 1000, 2000$ from the population. Each sampled dataset is analyzed independently, and performance measures are aggregated across Monte Carlo replications.

The simulation considers six data-generating processes designed to isolate different departures from the assumptions underlying classical ordered response models. Two benchmark settings generate latent errors from the normal and logistic distributions, under which ordered probit and ordered logit are correctly specified, respectively. The remaining settings introduce skewed latent errors, polarized (multimodal) latent attitudes, heteroskedastic latent errors, and nonlinear latent response functions. Together these scenarios represent empirical features frequently encountered in survey experiments while allowing the consequences of different modeling assumptions to be examined separately.

Five estimators are compared. The first is an empirical distribution estimator that estimates category probabilities directly from observed treatment-arm frequencies without imposing a statistical model. The second and third are ordered probit and ordered logit. The remaining estimators implement the proposed

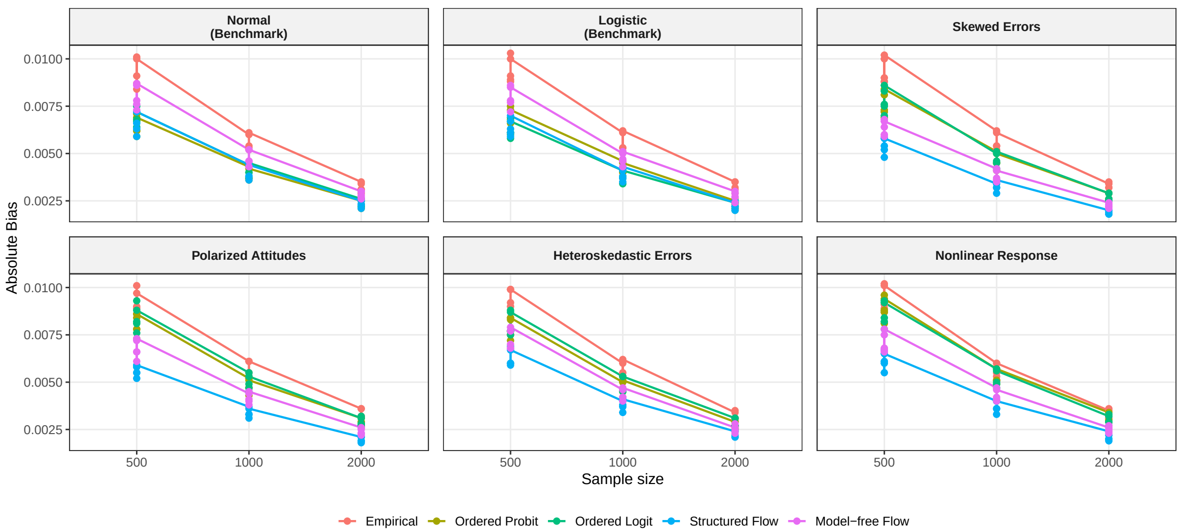

FIGURE 3

framework in two forms. The structured flow specification retains the linear latent-index model while estimating the latent error distribution flexibly. The model-free specification estimates both the systematic component and the latent error distribution nonparametrically. To facilitate stable optimization, both flow estimators are initialized using the ordered probit estimates before likelihood maximization.

Performance is evaluated using bias, absolute bias, and root mean squared error (RMSE) for the category-specific and cumulative treatment effects introduced in Section 2.3. These quantities are obtained by regression standardization from each fitted model. Because all estimators target the same causal estimands under identical identification assumptions, differences in performance reflect only the statistical consequences of the modeling assumptions imposed by each estimation procedure.

4.2. Simulation Results

Figures 3 and 4 summarize the finite-sample performance of the competing estimators across the six data-generating processes. Each panel corresponds to one latent response mechanism, while the horizontal axis reports the sample size. Absolute bias and root mean squared error (RMSE) are averaged across the category-specific and cumulative treatment effects introduced in Section 2.3. Lower values indicate more accurate recovery of the target causal quantities.

The benchmark settings provide an important validation of the proposed framework. Under normally distributed latent errors, where ordered probit is correctly specified, and under logistic latent errors, where ordered logit is correctly specified, the proposed estimators remain competitive with the classical models. In particular, the structured flow estimator exhibits little or no efficiency loss despite relaxing the para-

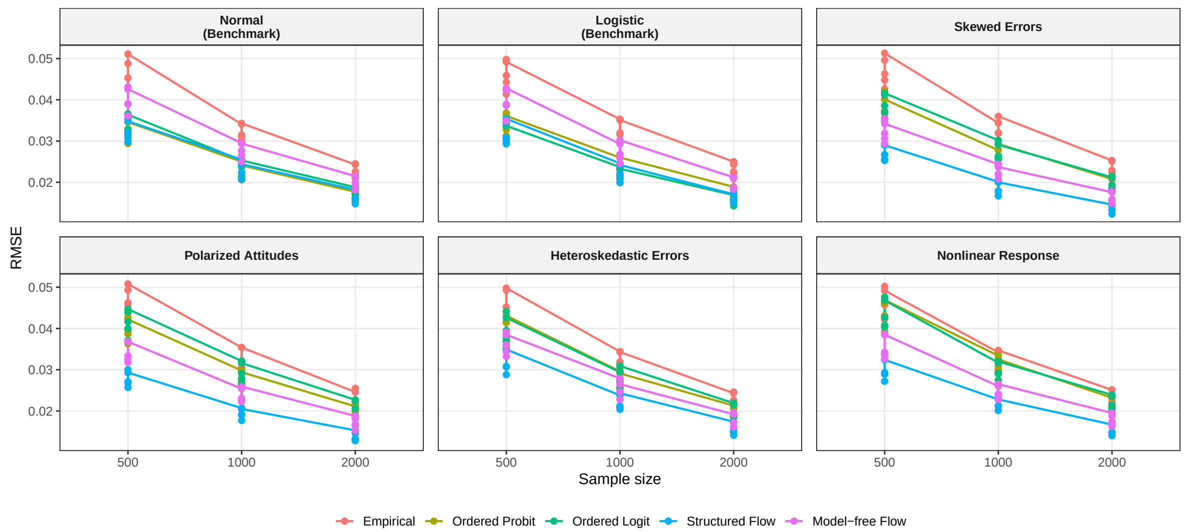

FIGURE 4

metric assumption on the latent error distribution. These benchmark scenarios therefore demonstrate that the additional modeling flexibility does not substantially compromise finite-sample performance when the conventional assumptions are approximately correct.

The remaining four scenarios examine departures from the assumptions underlying classical ordered response models. As the latent error distribution becomes skewed, multimodal, or heteroskedastic, or when the systematic component becomes nonlinear, the structured flow estimator consistently achieves the lowest estimation error among the model-based approaches. The gains are particularly pronounced under polarized latent attitudes and nonlinear response functions, where restrictive parametric assumptions lead to noticeable increases in both bias and RMSE for ordered logit and ordered probit. These results indicate that modeling the latent error distribution flexibly substantially improves recovery of the conditional ordinal response distribution when classical assumptions are violated.

The model-free flow estimator generally improves upon ordered logit and ordered probit under model misspecification but is typically outperformed by the structured flow specification. This pattern suggests that retaining the latent-index representation provides useful structural information while allowing flexibility in the latent error distribution. Conversely, the empirical distribution estimator exhibits the largest estimation error across nearly all settings because it estimates treatment-arm response probabilities directly from the observed sample without borrowing information across covariate values.

Across all simulation settings, estimation error decreases with sample size for every estimator, as expected. More importantly, the relative ranking of the estimators remains stable. The proposed framework therefore achieves the desired balance between robustness and efficiency: it remains competitive when the assumptions of classical ordered response models hold while providing substantial gains when those assumptions are violated.

5. Application 1: Tomz and Weeks (2020)

The first example replicates the analysis by Tomz and Weeks (2020). The study looked into the question of how the human rights practices of foreign adversaries affect domestic public support for war through survey experiments. The main argument was that in democracies with well-established human rights, people are less willing to go to war with other countries that also respect human rights than with comparable countries that violate them. Thus they are focusing on two factors that can impact people's support for war: the political system of the foreign country and how much the foreign country respect human rights.

The paper uses total of 5 surveys, but this paper focus on the first, which contains an experiment for their main result. The survey was conducted by YouGov with a nationally representative sample of 1,430 US adults in October 2012. The treatment was given as in a form of vignettes on a country that is developing nuclear weapons. The vignettes also include trade and military relationship of the country and the US, and its conventional military strength which is set to the half of that of the US. In the vignettes, information about the political system (democracy = 1, not = 0) and human rights practices (respect = 1, not = 0) were binary treatments and they were randomized independently.

The outcome of interest here is the support for physical strike of the US armed forces to the country. Specifically, respondents were asked to answer in 5-point scale: *Favor strongly, Favor somewhat, Neither favor nor oppose, Oppose somewhat*, and *Oppose strongly.* In the paper, the authors take the cardinalization approach and OLS. They first recode the outcome to take numeric values from 1 to 5, then they used this to run OLS. As discussed in Section 2, the cardinalization is arbitrary and we only can interpret the ATE as suggested by authors when the cardinalization is capturing the unknown transformation function, which is untestable.

The model they used in their main results (model 2 of their Table 1) can be expressed as following:

$$Y^* = \tau_{DEM} \cdot DEM + \tau_{HR} \cdot HR + X^T \beta + \varepsilon$$

$$Y \in \{1, 2, 3, 4, 5\}$$

where Y^* is the support for the military strike in latent scale and Y is the observed outcome with cardinalization from 1 to 5, *DEM* and *HR* are binary variables indicating political system treatment and human rights practices treatment respectively, X for the control variables and ε for the error term.

Since they are interested in whether citizens living in democracy where human rights are respected (the US, to be more specific) are more reluctant to support military strike to a country who is developing nuclear weapons if the country also respecting human rights or the country has democratic political system, the causal parameters we are interested in the above equation are τ_{HR} and τ_{DEM}. We only can observe the ordinal outcome, we can only identify the effects up to scale, and here I adopted error-variance normalization we discussed in the previous section. I replicate their result on Table S3 of their paper, and use ordered logit, ordered probit and KDE-based estimator as a comparison.

	ATE	NLTE			
	Original – OLS	Ordered Logit	Ordered Probit	KDE-based	NF-based
Respect HR	-12.6^{***}	-0.46^{***}	-0.49^{***}	-0.48	-0.49^{***}
	(1.47)	(0.10)	(0.06)	(0.27)	(0.06)
Democracy	-9.5^{***}	-0.31^{**}	-0.34^{***}	-0.33	-0.34^{***}
	(1.47)	(0.10)	(0.06)	(0.22)	(0.06)

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

TABLE 1. Replication of Tomz and Weeks (2020) Table S3. ATE represents the original binarized OLS estimates (0-100 scale). NLTE columns report standardized latent coefficients ($\sigma = 1$).

Table 1 reports the results across all five specifications. Substantively, all models agree on the direction of the treatment effects: information that a country respects human rights or is a democracy significantly decreases domestic support for military action. However, the interpretation and statistical information utilized differ. The original OLS interpretation (a 12.6 percentage point drop for the Human Rights treatment) relies on the validity of the binarization and the cardinal 0–100 scale. By collapsing the 5-point scale into two categories, the original analysis discards potentially valuable information regarding the intensity of respondent preferences.

In contrast, NLTE models utilize the full ordinal range of the data. The standardized coefficients remain remarkably consistent across the Ordered Probit, KDE-based, and NF-based models. For instance, the effect of respecting human rights is estimated at approximately -0.49 standard deviations across these specifications. The fact that the flexible semiparametric estimators (KDE and NF) closely track the Ordered Probit results serves as a diagnostic, suggesting that the latent error distribution for this specific survey approximates a normal distribution. In this instance, the parametric assumption appears robust, though the semiparametric approach provides the necessary validation that the original cardinalization was not distorting the underlying direction of the effects.

6. Application 2: Mattingly et al. (2025)

The second application replicates the large-scale cross-national study by Mattingly et al. (2025), which examines how authoritarian regimes utilize state-produced media to promote their political and economic models to foreign audiences. We focus our replication on the authors' primary hypotheses: (H1) US government messaging increases preference for the American model, (H2) Chinese Communist Party (CCP) messaging increases preference for the Chinese model, and (H3) in the presence of competing messages, public preference shifts toward the Chinese model.

To test the hypothesis, authors launched multiple survey experiments took place in 19 different countries (N = 6,000) in June 2022. Respondents were assigned via block randomization to one of four conditions: viewing US state-produced videos (USA), CCP-produced videos (China), a combination of both (Competition), or nature-themed placebo videos (Control). The primary outcomes were measured using a 6-point Likert scale, ranging from *Strongly Prefer the US* (1) to *Strongly Prefer China* (6). The original analysis relied on OLS regression, which cardinalizes these ordinal categories by treating the 1–6 scale as an interval measure.

After the treatment, respondents were asked to answer two questions which measures main outcome variables (whether they prefer the US (political / economic) system or (political / economic) Chinese system) in 6 categories from *Strongly Prefer the US* to *Strongly Prefer China*. The cardinalize the outcomes by assigning numeric values, so that *Strongly Prefer the US* was recoded as 1 and *Strongly Prefer China* was recoded as 6. The original results came from OLS results with these cardinalized outcomes.

The model Mattingly et al. (2025) used for their main results can be expressed as follows

$$Y_i^* = \tau_{CHINA} \cdot CHINA_i + \tau_{USA} \cdot USA_i + \tau_{COMP} \cdot COMP_i + X_i^T \gamma + \varepsilon_i$$

$$Y \in \{1, 2, 3, 4, 5, 6\}$$

where $CHINA_i$, USA_i and $COMP_i$ are indicator variables for the respective experimental conditions, and X_i is a vector of pre-treatment covariates including age, gender, education and family income.

While the original study estimates the Average Treatment Effect (ATE) by applying OLS directly to the categorical labels (1–6), our approach focuses on identifying the latent treatment parameters (τ). As is standard in semiparametric ordinal models, the latent scale is identified only up to location and scale; hence, we apply the error-variance normalization ($\sigma = 1$). This allows for a direct comparison between the parametric specifications (Ordered Probit and Logit) and the flexible KDE and NF-based estimators. Under this framework, the coefficients represent the shift in the latent preference distribution in units of standard deviations, providing a measure of treatment intensity that is robust to the arbitrary cardinalization of the 6-point scale.

Table 2 reports the replication of the original OLS results, and results of NLTE estimation using different ordinal regression models including KDE-based model and flow-based model. For preferences over the political system (upper panel), we find a stark divergence between parametric and semiparametric models. While the direction of the effects remains consistent across all specifications, the magnitude of the estimates is significantly inflated in the cardinal (OLS) and parametric ordinal (Logit/Probit) models. Specifically, the estimated effect of the Chinese treatment in the NF-based model (0.37) is nearly two-thirds smaller than the OLS estimate (1.04) and less than half the size of the Ordered Probit estimate (0.76).

Notably, the *Competition* effect, which the original study identified as a robust driver of preference for the Chinese model (0.36, $p < 0.001$), is substantially attenuated in the flexible models. In the NF-based specification, this effect drops to 0.11, suggesting that when the assumption of a Normal error distribution is relaxed, the persuasive power of competitive rhetoric appears much more fragile than previously reported. This suggests that the OLS and Probit models may be over-attributing shifts in the latent distribution to the treatment, rather than accounting for the non-normal dispersion of political preferences.

In contrast, the lower panel of Table 2 reports preferences over economic systems, where the models show greater consensus. While OLS still yields the largest point estimates, the gap between the Probit (0.57) and the NF-based model (0.58) for the China treatment is negligible. This indicates that the latent distribution of economic preferences in this sample likely approximates normality more closely than political preferences. However, the OLS estimates for all treatments remain consistently higher than their latent counterparts, reinforcing the argument that cardinalizing Likert scales may lead to substantially different conclusions.

7. Conclusion

This paper has examined the problem of causal inference when the outcome of interest is measured on an ordinal scale, a situation that arises frequently in political science research on attitudes and preferences. Building on a standard latent-variable framework, we showed that, even under conventional causal assumptions such as SUTVA, ignorability, and positivity, treatment effects on the underlying continuous outcome are in general identified only up to scale. The joint distribution of ordinal responses is invariant to positive affine transformations of the latent outcome and thresholds, so the absolute magnitude and location of the latent average treatment effect cannot be recovered from the data alone. Any attempt to compare coefficients (across models or across studies) must therefore adopt an explicit normalization. To address this, we

Outcome: Preference over Political System					
	ATE	NLTE			
	Original – OLS	Ordered Logit	Ordered Probit	KDE-based	NF-based
China	1.04***	0.73***	0.76***	0.36***	0.37***
	(0.05)	(0.04)	(0.04)	(0.07)	(0.04)
USA	−0.43***	−0.35***	−0.38***	−0.18***	−0.17***
	(0.04)	(0.04)	(0.04)	(0.05)	(0.03)
Competition	0.36***	0.21***	0.25***	0.13*	0.11**
	(0.05)	(0.04)	(0.04)	(0.06)	(0.04)
Outcome: Preference over Economic System					
	ATE	NLTE			
	Original – OLS	Ordered Logit	Ordered Probit	KDE-based	NF-based
China	0.87***	0.56***	0.57***	0.59***	0.58***
	(0.05)	(0.04)	(0.04)	(0.07)	(0.06)
USA	−0.57***	−0.39***	−0.43***	−0.42***	−0.43***
	(0.05)	(0.04)	(0.04)	(0.05)	(0.05)
Competition	0.28***	0.16***	0.18***	0.19**	0.18**
	(0.06)	(0.04)	(0.04)	(0.06)	(0.06)

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

TABLE 2. Replication of Mattingly et al. (2025) Table C3. ATE represents original OLS estimates. NLTE columns report standardized latent coefficients from various ordinal specifications.

proposed the *Normalized Latent Treatment Effect (NLTE)* as an alternative causal estimand, ensuring that latent treatment effects are expressed in a transparent and comparable metric.

We argued that common strategies for analyzing ordinal outcomes are problematic in light of this limitation. Cardinalization, which assigns numeric scores to categories and applies tools such as OLS or difference-in-means, produces estimates whose size and sometimes even sign can depend sensitively on arbitrary labeling choices (Schröder and Yitzhaki, 2017; Bond and Lang, 2018; Bloem, 2022). Binarization discards information about intermediate categories, leading to inefficient estimators and potentially obscuring substantively important changes in the distribution of responses. Parametric ordinal models, including ordered logit and ordered probit, avoid arbitrary numeric labels and respect the ordered nature of the data, but rely on strong assumptions about the latent error distribution. When these assumptions are violated the resulting estimators converge to pseudo-true parameters and can suffer substantial bias (Manski, 1988; Greene and Hensher, 2010; Johnston, McDonald and Quist, 2020; Smits et al., 2020).

This paper introcused two estimators that retain the interpretability of latent treatment effects while relaxing parametric assumptions on the error distribution. The first is a semiparametric KDE-based estimator that uses kernel density estimation to approximate the latent error CDF and constructs a quasi-likelihood for the ordinal model. The second employs normalizing flows to model the error distribution as an invertible transformation of a simple base distribution within a maximum likelihood framework. Both estimators treat the latent outcome as parametric, estimate the error distribution flexibly from the data.

Monte Carlo simulations demonstrated that these choices matter in practice. When the latent error distribution is correctly specified (for example, normal for ordered probit), parametric ordinal models perform well, as expected. However, under misspecification such as lognormal error, ordered logit and probit can be noticeably biased, whereas the KDE-based and NF-based estimators remain consistent and exhibit favorable finite-sample performance. The simulations also showed that cardinalization yields highly unstable estimates across different labeling schemes, and that binarization is less efficient and can miss meaningful shifts among categories. In this sense, the density-estimation-based approaches provide a robust alternative that better respects the ordinal nature of the data and the limited information it contains about the latent scale.

The empirical applications further underscore the substantive importance of these methods. Reanalyzing Tomz and Weeks (2020) on human rights and military force, we found that standard binarized OLS tends to understate the magnitude of the latent treatment effect. Conversely, our replication of Mattingly et al. (2025) regarding authoritarian propaganda reveals that the perceived "superiority" of Chinese state cues over American cues in the original study is, in part, a methodological artifact. By accounting for the floor effects and non-normal dispersion of political preferences, our proposed estimators show that the relative persuasive power of these cues is much closer than cardinal OLS suggests. These examples demonstrate how the assumption of cardinality or normality can inadvertently shape—and potentially distort—substantive conclusions in top-tier political science research.

Several directions for future work follow from these results. First, while we focused on single-item ordinal outcomes, extending density-estimation-based methods to multi-item settings and integrating them more tightly with IRT-style measurement models would be valuable. Second, developing practical diagnostic tools for assessing error distribution misspecification in ordinal models, building on surrogate residuals or related techniques, would complement the estimators proposed here. Third, applications with clustered, panel, or hierarchical data structures pose additional challenges for both identification and estimation that merit further study. More broadly, the analysis reinforces the importance of taking the ordinal nature of many political science outcomes seriously and of being explicit about the assumptions and normalizations that underlie estimates of causal effects on latent attitudes.

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8. Appendix

A. KDE-Based Ordinal Regression

This appendix provides additional details for the semiparametric KDE-based estimator introduced in Section 3. We first define the quasi-likelihood, then describe the kernel estimation of the error distribution and state a consistency result.

A.1. Model and Quasi-Likelihood

Recall the latent outcome model

$$Y_i^* = f(X_i, \beta) + D_i^\top \tau + \varepsilon_i,$$

and the threshold-based reporting function

$$Y_i = j \iff \alpha_{j-1} < Y_i^* \le \alpha_j, \qquad j = 0, 1, \ldots, J,$$

with $-\infty = \alpha_{-1} < \alpha_0 < \cdots < \alpha_J = \infty$. For notational simplicity, define the latent outcome

$$V_i(\beta, \tau) = f(X_i, \beta) + D_i^\top \tau.$$

Let F denote the (unknown) CDF of ε_i. Conditional on (X_i, D_i), the probability of observing category j is

$$p_j(X_i, D_i; \alpha, \beta, \tau, F) = F\big(\alpha_j - V_i(\beta, \tau)\big) - F\big(\alpha_{j-1} - V_i(\beta, \tau)\big).$$

If F were known, the population log-likelihood would be

$$\ell(\alpha, \beta, \tau; F) = \mathbb{E}\left[\sum_{j=0}^{J} 1_{Y_i = j} \log p_j(X_i, D_i; \alpha, \beta, \tau, F)\right].$$

In the sample of size n, the infeasible sample log-likelihood is

$$\ell_n(\alpha, \beta, \tau; F) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=0}^{J} 1_{Y_i = j} \log p_j(X_i, D_i; \alpha, \beta, \tau, F).$$

When F is unknown, we replace it by a nonparametric estimator $\hat{F}$ constructed via KDE, obtaining a *quasi*-log-likelihood

$$\hat{\ell}_n(\alpha, \beta, \tau) = \frac{1}{n} \sum_{i=1}^{n} \hat{m}(X_i) \sum_{j=0}^{J} 1_{Y_i = j} \log \hat{p}_j(X_i, D_i; \alpha, \beta, \tau),$$

where

$$\hat{p}_j(X_i, D_i; \alpha, \beta, \tau) = \hat{F}\big(\alpha_j - V_i(\beta, \tau)\big) - \hat{F}\big(\alpha j - 1 - V_i(\beta, \tau)\big),$$

and $\hat{m}(X_i)$ is a trimming function that downweights observations in extreme regions of the covariate space where KDE may be unstable. The KDE-based estimator $(\hat{\alpha}, \hat{\beta}, \hat{\tau})$ is defined as any maximizer of $\hat{\ell}_n(\alpha, \beta, \tau)$.

A.2. Kernel Estimation of the Error Distribution

To construct $\hat{F}$, we exploit the relationship between the distribution of the index V_i and the error term ε_i. Let

$$g_1(v \mid Y \le j) = \text{density of } V \text{ given } Y \le j, \qquad g_0(v \mid Y > j) = \text{density of } V \text{ given } Y > j,$$

and let

$$\pi_j = \mathbb{P}(Y_i \le j), \qquad 1 - \pi_j = \mathbb{P}(Y_i > j).$$

Then by the Bayes' rule,

$$\mathbb{P}(Y_i \le j \mid V_i = v) = \frac{\pi_j g_1(v \mid Y \le j)}{\pi_j g_1(v \mid Y \le j) + (1 - \pi_j) g_0(v \mid Y > j)}.$$

Since $Y_i^* = V_i + \varepsilon_i$, and $Y_i \le j$ iff $Y_i^* \le \alpha_j$, we have

$$\mathbb{P}(Y_i \le j \mid V_i = v) = \mathbb{P}\big(\varepsilon_i \le \alpha_j - v\big) = F\big(\alpha_j - v\big).$$

Thus, if we can estimate π_j, $g_1(\cdot)$, and $g_0(\cdot)$ from the data, we can construct an estimator $\hat{F}$ such that

$$\hat{F}(\alpha_j - v) \approx \widehat{\mathbb{P}}(Y_i \le j \mid V_i = v).$$

The probabilities π_j and $1 - \pi_j$ can be estimated by sample proportions:

$$\hat{\pi}_j = \frac{1}{n} \sum_{i=1}^{n} 1_{Y_i \le j}, \qquad 1 - \hat{\pi}_j = \frac{1}{n} \sum_{i=1}^{n} 1_{Y_i > j}.$$

The conditional densities g_1 and g_0 are estimated using KDE. For concreteness, suppose we use a univariate kernel $K(\cdot)$ (e.g. Gaussian) and bandwidths that may depend on j. Let $n_1(j) = \sum_{i=1}^{n} 1_{Y_i \le j}$ and $n_0(j) =$

$\sum_{i=1}^{n} 1_{Y_i > j}$. Define the subsamples

$$V_i : Y_i \leq j, \qquad V_i : Y_i > j,$$

and estimate the conditional densities as

$$\hat{g}_1(v \mid Y \leq j) = \frac{1}{n_1(j) h_{1j}} \sum_{i:\, Y_i \leq j} K\left(\frac{v - V_i}{h_{1j}}\right),$$

$$\hat{g}_0(v \mid Y > j) = \frac{1}{n_0(j) h_{0j}} \sum_{i:\, Y_i > j} K\left(\frac{v - V_i}{h_{0j}}\right),$$

where h_{1j} and h_{0j} are bandwidths chosen according to standard rules (possibly location-adaptive). Then the estimated conditional probability is

$$\widehat{\mathbb{P}}(Y_i \leq j \mid V_i = v) = \frac{\hat{\pi}_j \hat{g}_1(v \mid Y \leq j)}{\hat{\pi}_j \hat{g}_1(v \mid Y \leq j) + (1 - \hat{\pi}_j) \hat{g}_0(v \mid Y > j)}.$$

We interpret this as an estimator of the CDF of ε_i evaluated at $\alpha_j - v$:

$$\hat{F}(\alpha_j - v) \equiv \widehat{\mathbb{P}}(Y_i \leq j \mid V_i = v).$$

In practice, the KDE is implemented with local smoothing and trimming to reduce boundary bias and instability in regions with sparse data. Following Abramson (1982) and Silverman (1986), one can use pilot density estimates to define location-specific bandwidths and damping functions that prevent the bandwidth from shrinking too quickly in low-density regions. We adopt these techniques, as in Klein and Sherman (2002), to ensure $\sqrt{n}$ consistency.

A.3. Consistency of the KDE-Based Estimator

This section sketches the main consistency result for the KDE-based estimator. Fuller proofs follow from standard arguments combined with the consistency of locally smoothed KDEs (Klein and Spady, 1993; Abramson, 1982; Silverman, 1986).

Assumption 1 (Smoothness of the Index Distribution). *The conditional density of the index $V_i(\beta, \tau)$ given (X_i, D_i) exists, is strictly positive on its support, and is continuously differentiable up to a sufficiently high order. The support of V_i is compact or can be truncated to a compact set via trimming without affecting the asymptotic behavior of the estimator.*

Assumption 2 (KDE Regularity Conditions). *The kernel* $K(\cdot)$ *is a bounded, symmetric density with compact support. Bandwidths* h_{1j} *and* h_{0j} *satisfy* $h_{1j} \to 0$, $h_{0j} \to 0$, *and* $nh_{1j} \to \infty$, $nh_{0j} \to \infty$ *as* $n \to \infty$, *with additional conditions ensuring the consistency of locally smoothed KDEs (e.g. rates controlling the pilot bandwidth and damping parameters).*

Under these conditions, the KDE estimators $\hat{g}_1$ and $\hat{g}_0$ converge uniformly to g_1 and g_0, and the estimated CDF $\hat{F}$ converges uniformly to the true F on compact subsets of the support. Consequently, the quasi-log-likelihood $\hat{\ell}_n(\alpha, \beta, \tau)$ converges uniformly to $\ell_n(\alpha, \beta, \tau; F)$, and the maximizer of $\hat{\ell}_n$ converges to the maximizer of the infeasible log-likelihood (up to the usual positive affine transformation).

Theorem 2 (Consistency of the KDE-Based Estimator). *Suppose the latent outcome model and threshold structure are correctly specified, and Assumptions 1–2 hold. Then, as* $n \to \infty$,

$$\left|(\hat{\alpha}, \hat{\beta}, \hat{\tau}) - (\alpha, \beta, \tau)\right| = o_p(1),$$

up to an arbitrary positive affine transformation $(a + c\alpha, c\beta, c\tau)$ *with* $c > 0$. *In particular, the latent treatment effects* τ *are consistently estimated up to scale.*

Because the ordinal model is only identified up to scale, we impose the error-variance normalization $\mathrm{Var}(\varepsilon_i) = 1$ ex post by rescaling the coefficients using the estimated error variance $\hat{\sigma}_\varepsilon^2$ as described in Section 3. Under this normalization, the KDE-based estimator yields treatment effects that are directly comparable to those from ordered probit, ordered logit (after rescaling), and the normalizing-flow-based estimator.

B. Normalizing Flows

This appendix provides additional detail on the normalizing-flow-based estimator described in Section 3. We briefly review the change-of-variables formula, define the ordinal likelihood with a learned error distribution, and state a consistency result.

B.1. Change-of-Variables Formula

Let Z be a random variable with a known base density $f_Z(z)$ (for example, the standard normal density), and let $T_\theta : \mathbb{R} \to \mathbb{R}$ be an invertible, differentiable transformation parameterized by θ. Define

$$\varepsilon = T_\theta(Z).$$

Then, by the change-of-variables formula, the density of ε is

$$f_\theta(\varepsilon) = f_Z\big(T_\theta^{-1}(\varepsilon)\big)\left|\frac{d}{d\varepsilon}T_\theta^{-1}(\varepsilon)\right|.$$

The corresponding CDF F_θ is

$$F_\theta(u) = \int_{-\infty}^{u} f_\theta(t), dt.$$

We refer to the family $T_{\theta\theta\in\Theta}$ as a *normalizing flow*. By choosing a sufficiently flexible family (e.g. coupling layers, spline flows), we can approximate a wide class of univariate distributions with tractable densities and gradients.

B.2. Ordinal Likelihood with a Flow-Based Error Distribution

We embed the flow-based error distribution into the latent outcome model,

$$Y_i^* = f(X_i, \beta) + D_i^\top \tau + \varepsilon_i, \qquad \varepsilon_i = T_\theta(Z_i), \quad Z_i \sim \mathcal{N}(0, 1),$$

with thresholds α defining the observed ordinal outcome:

$$Y_i = j \quad \Longleftrightarrow \quad \alpha_{j-1} < Y_i^* \le \alpha_j.$$

Let f_θ and F_θ denote the density and CDF of ε_i implied by the flow T_θ. Conditional on (X_i, D_i), the probability of category j is

$$\mathbb{P}(Y_i = j \mid X_i, D_i; \alpha, \beta, \tau, \theta) = F_\theta\big(\alpha_j - V_i(\beta, \tau)\big) - F_\theta\big(\alpha_{j-1} - V_i(\beta, \tau)\big),$$

where $V_i(\beta, \tau) = f(X_i, \beta) + D_i^\top \tau$. The sample log-likelihood is

$$\ell_n(\alpha, \beta, \tau, \theta) = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=0}^{J} 1_{Y_i = j} \log\left[F_\theta\big(\alpha_j - V_i(\beta, \tau)\big) - F_\theta\big(\alpha_{j-1} - V_i(\beta, \tau)\big)\right].$$

We estimate $(\alpha, \beta, \tau, \theta)$ jointly by maximizing $\ell_n(\alpha, \beta, \tau, \theta)$ with respect to all parameters. Because T_θ is invertible and differentiable, both f_θ and F_θ are tractable, and gradients of the log-likelihood with respect to $(\alpha, \beta, \tau, \theta)$ can be computed efficiently using automatic differentiation.

B.3. Consistency of the Flow-Based Estimator

This section briefly states a consistency result for the Normalizing-Flow-Based estimator. Let $(\alpha_0, \beta_0, \tau_0, F_0)$ denote the true parameters and error distribution. Suppose that the true error distribution F_0 belongs to the closure of the flow family $F_\theta : \theta \in \Theta$, and that the latent outcome model and threshold structure are correctly specified.

Assumption 3 (Flow Approximation and Identification). *The flow family $F_\theta : \theta \in \Theta$ is rich enough that there exists $\theta_0 \in \Theta$ such that $F_{\theta_0} = F_0$ (or F_{θ_0} approximates F_0 arbitrarily well). The parameter vector $(\alpha, \beta, \tau, \theta)$ is identified up to a positive affine transformation $(a + c\alpha, c\beta, c\tau)$ with $c > 0$.*

Assumption 4 (Regularity Conditions). *The parameter space for $(\alpha, \beta, \tau, \theta)$ is compact or can be restricted to a compact subset; the log-likelihood function $\ell_n(\alpha, \beta, \tau, \theta)$ satisfies standard regularity conditions (continuity, differentiability, integrability); and the model is correctly specified in the sense that the true data-generating process is in the model class for some $(\alpha_0, \beta_0, \tau_0, \theta_0)$.*

Under these conditions, standard maximum likelihood theory implies that the flow-based estimator is consistent (up to scale) and asymptotically normal.

Theorem 3 (Consistency of the Flow-Based Estimator). *Suppose the latent outcome model and threshold structure are correctly specified, and Assumptions 3–4 hold. Then, as $n \to \infty$,*

$$\left|(\hat{\alpha}, \hat{\beta}, \hat{\tau}, \hat{\theta}) - (\alpha_0, \beta_0, \tau_0, \theta_0)\right| = o_p(1),$$

up to an arbitrary positive affine transformation $(a + c\alpha_0, c\beta_0, c\tau_0)$ with $c > 0$. In particular, the latent treatment effects τ_0 are consistently estimated up to scale.

As with the KDE-based estimator, we impose the error-variance normalization $\mathrm{Var}(\varepsilon_i) = 1$ ex post by rescaling the coefficients using the estimated error variance. Given $\hat{\theta}$, we estimate

$$\hat{\sigma}\varepsilon^2 = \mathrm{Var}\hat{F}\theta(\varepsilon_i) \approx \frac{1}{M} \sum m = 1^M \left(T_{\hat{\theta}}(Z_m)\right)^2,$$

where $Z_m \sim \mathcal{N}(0, 1)$ are independent draws from the base distribution. We then report

$$\hat{\tau}^{\text{unit-var}} = \frac{\hat{\tau}}{\hat{\sigma}\varepsilon}, \qquad \hat{\beta}^{\text{unit-var}} = \frac{\hat{\beta}}{\hat{\sigma}\varepsilon},$$

so that all treatment effects are expressed on the same unit-error-variance latent scale as in the main text.